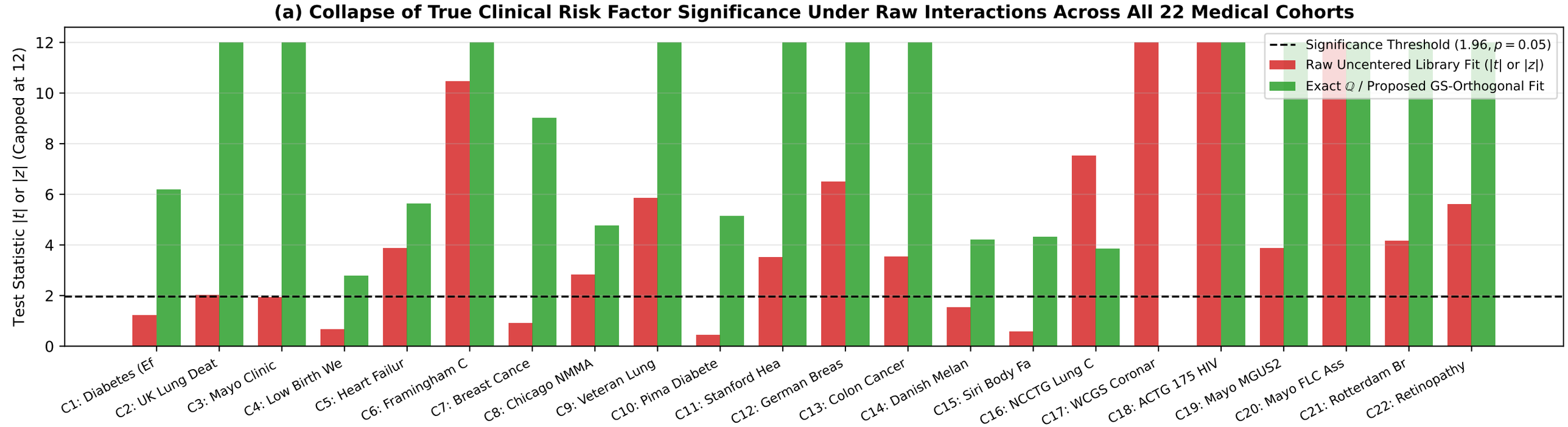
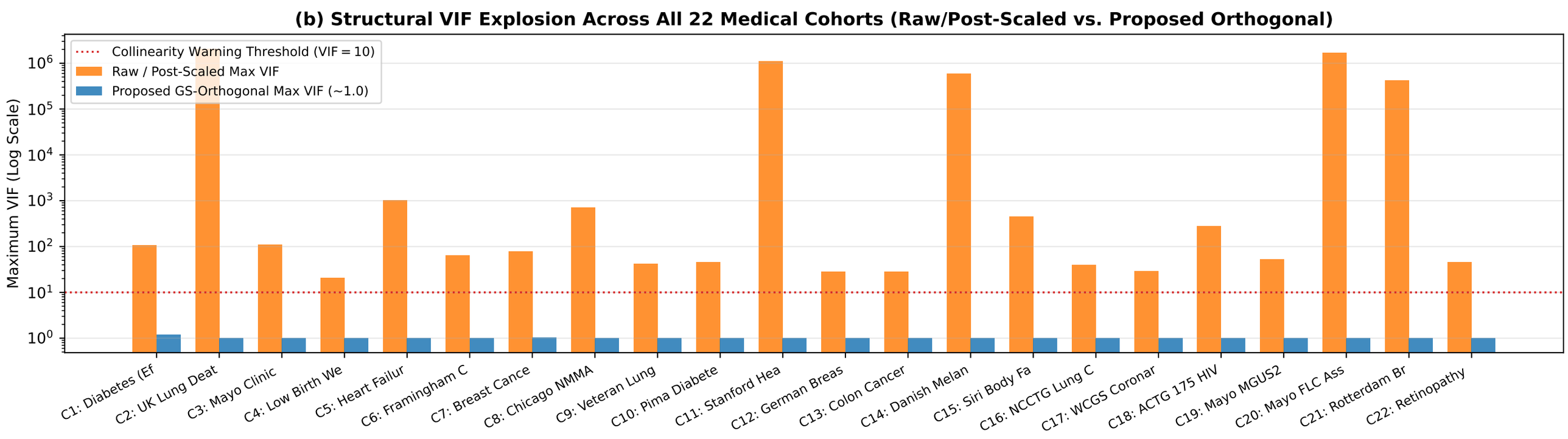

(a) Collapse of True Clinical Risk Factor Significance Under Raw Interactions Across All 22 Medical Cohorts
Test Statistic |t| or |z| (Capped at 12)
Significance Threshold (1.96, p = 0.05)
Raw Uncentered Library Fit (|t| or |z|)
Exact 𝒬 / Proposed GS-Orthogonal Fit
0
2
4
6
8
10
12
C1: Diabetes (Ef
C2: UK Lung Deat
C3: Mayo Clinic
C4: Low Birth We
C5: Heart Failur
C6: Framingham C
C7: Breast Cance
C8: Chicago NMMA
C9: Veteran Lung
C10: Pima Diabete
C11: Stanford Hea
C12: German Breas
C13: Colon Cancer
C14: Danish Melan
C15: Siri Body Fa
C16: NCCTG Lung C
C17: WCGS Coronar
C18: ACTG 175 HIV
C19: Mayo MGUS2
C20: Mayo FLC Ass
C21: Rotterdam Br
C22: Retinopathy


(b) Structural VIF Explosion Across All 22 Medical Cohorts (Raw/Post-Scaled vs. Proposed Orthogonal)
Maximum VIF (Log Scale)
Collinearity Warning Threshold (VIF = 10)
Raw / Post-Scaled Max VIF
Proposed GS-Orthogonal Max VIF (~1.0)
10^0
10^1
10^2
10^3
10^4
10^5
10^6
C1: Diabetes (Ef
C2: UK Lung Deat
C3: Mayo Clinic
C4: Low Birth We
C5: Heart Failur
C6: Framingham C
C7: Breast Cance
C8: Chicago NMMA
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C19: Mayo MGUS2
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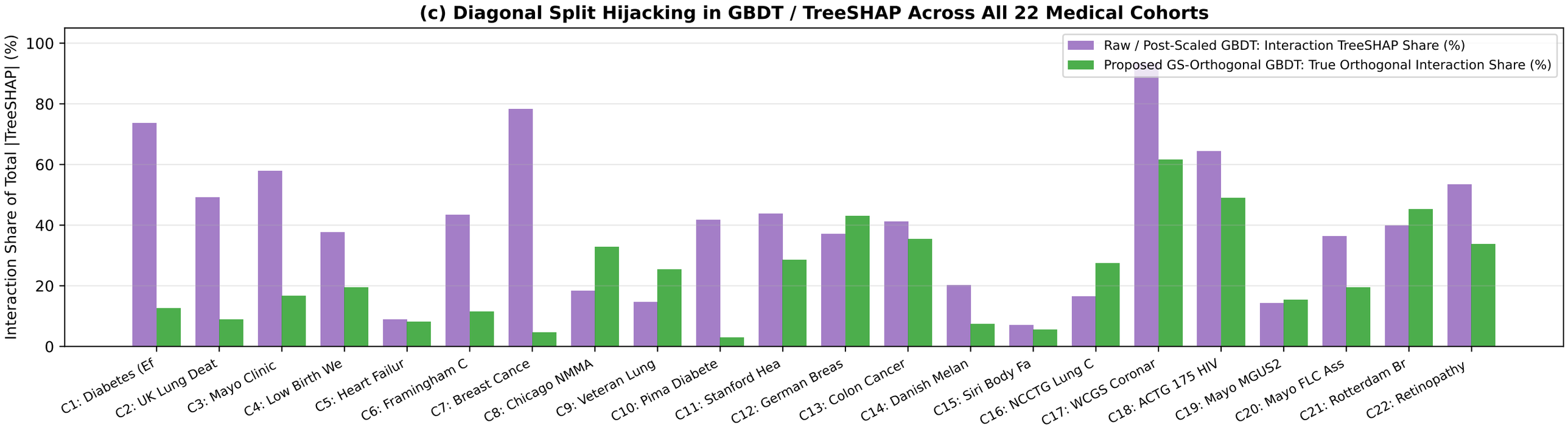
(c) Diagonal Split Hijacking in GBDT / TreeSHAP Across All 22 Medical Cohorts
Interaction Share of Total |TreeSHAP| (%)
Raw / Post-Scaled GBDT: Interaction TreeSHAP Share (%)
Proposed GS-Orthogonal GBDT: True Orthogonal Interaction Share (%)
0
20
40
60
80
100
C1: Diabetes (Ef
C2: UK Lung Deat
C3: Mayo Clinic
C4: Low Birth We
C5: Heart Failur
C6: Framingham C
C7: Breast Cance
C8: Chicago NMMA
C9: Veteran Lung
C10: Pima Diabete
C11: Stanford Hea
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C13: Colon Cancer
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C21: Rotterdam Br
C22: Retinopathy